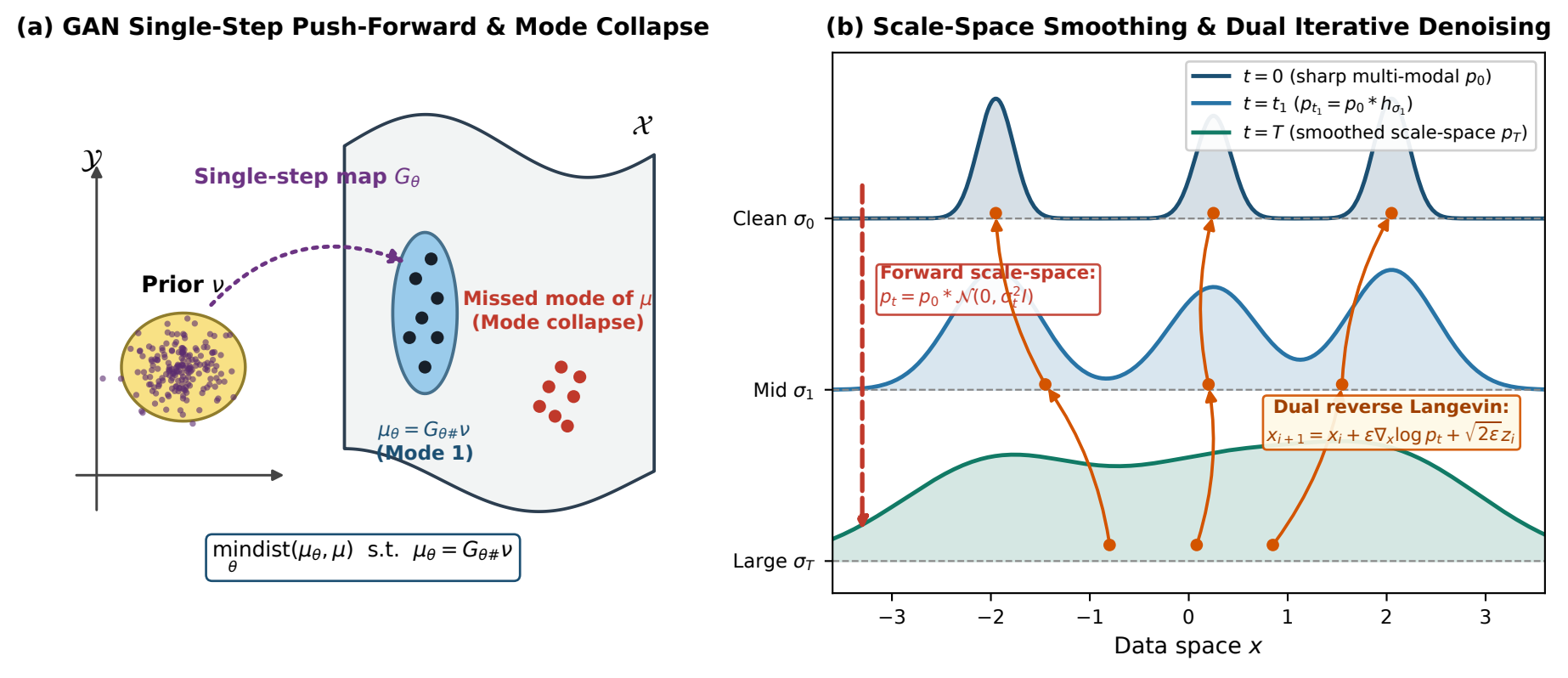

(a) GAN Single-Step Push-Forward & Mode Collapse
$\mathcal{Y}$
$\mathcal{X}$
Single-step map G_θ
Prior ν
Missed mode of μ
(Mode collapse)
$\mu_\theta = G_{\theta\#}\nu$
(Mode 1)
$\min_\theta \text{dist}(\mu_\theta, \mu)$ s.t. $\mu_\theta = G_{\theta\#}\nu$
(b) Scale-Space Smoothing & Dual Iterative Denoising
$t = 0$ (sharp multi-modal p_0)
$t = t_1$ ($p_{t_1} = p_0 * h_{\sigma_1}$)
$t = T$ (smoothed scale-space p_T)
Clean σ_0
Mid σ_1
Large σ_T
Forward scale-space:
$p_t = p_0 * \mathcal{N}(0, \sigma_t^2 I)$
Dual reverse Langevin:
$x_{i+1} = x_i + \varepsilon \nabla_x \log p_t + \sqrt{2\varepsilon} z_i$
−3
−2
−1
0
1
2
3
Data space x